

Sampling and reconstruction

Zhuoqi Cheng

zch@mmmi.sdu.dk

SDU Robotics

Topics to be covered in this course

→ Sampling and reconstruction

→ Aliasing

→ Quantization and dynamic range

→ Implementation

→ Conversion time-frequency domain

→ Z transform

→ Linear Time Invariant system (LTI)

→ System analysis

→ Window functions

→ Filter design

→ Impulse response (FIR and IIR)

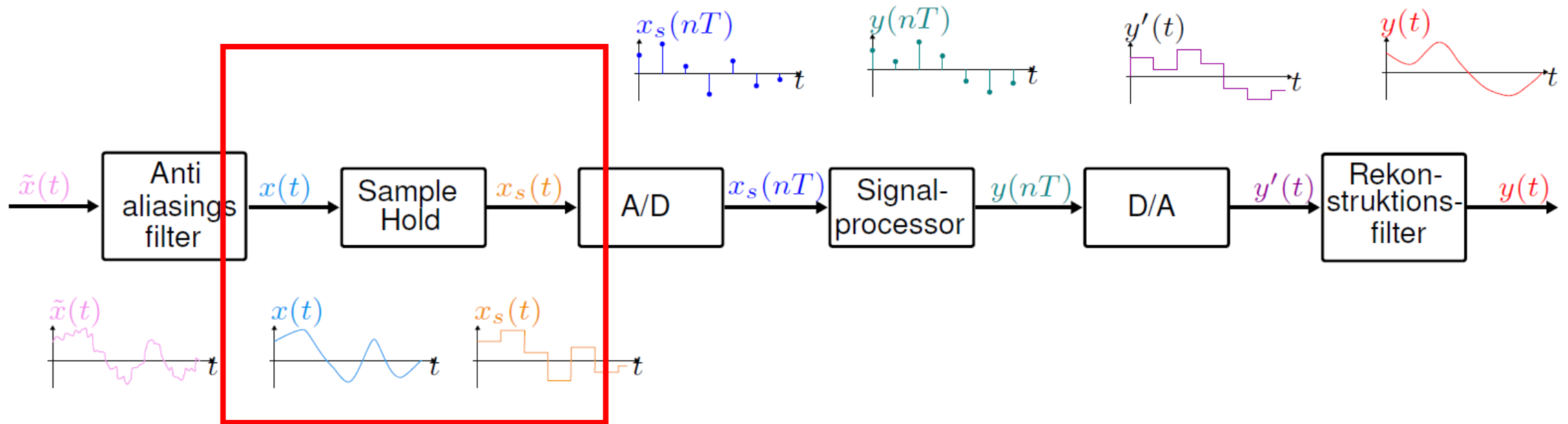
Signal Sampling and Quantization

2

Reference book

Chapter 2: Signal Sampling and Quantization
(Page 15-47)

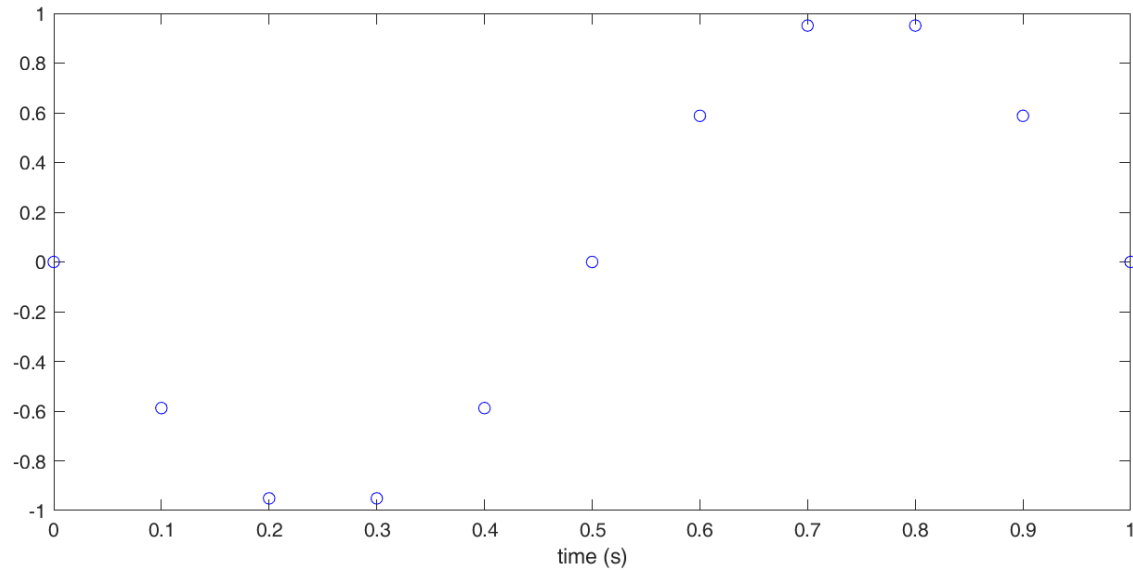
Digital signal processing workflow



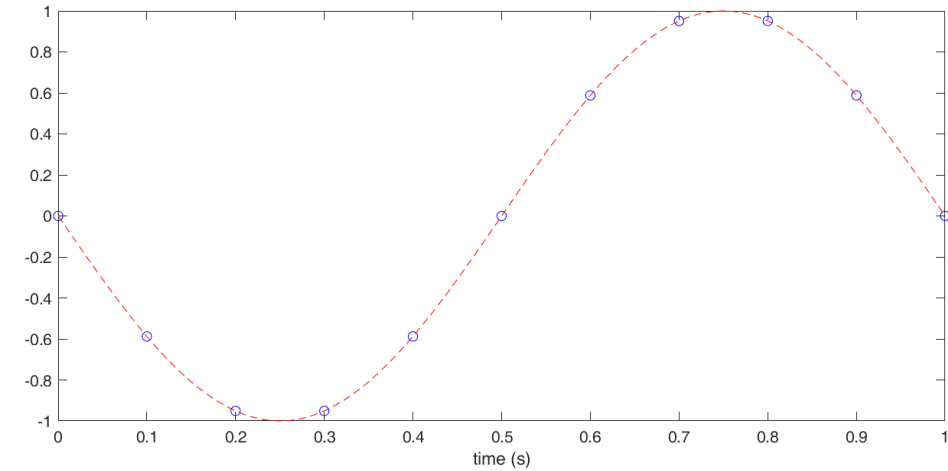
Question?

The input discrete signal is as follows. Assuming that it is a sinusoidal signal.

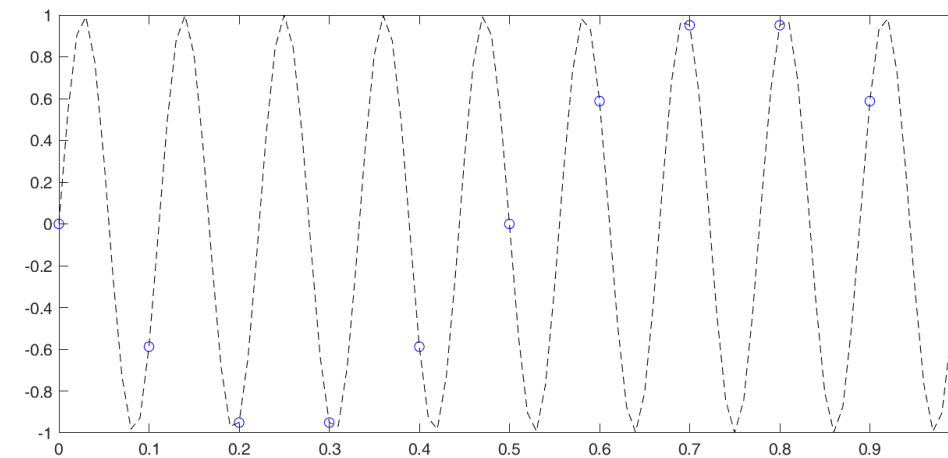
Q1: what is the sampling frequency f_s ?



Q2: what is your guess of the signal's frequency?



$f=1\text{Hz}$



$f=9\text{Hz}$

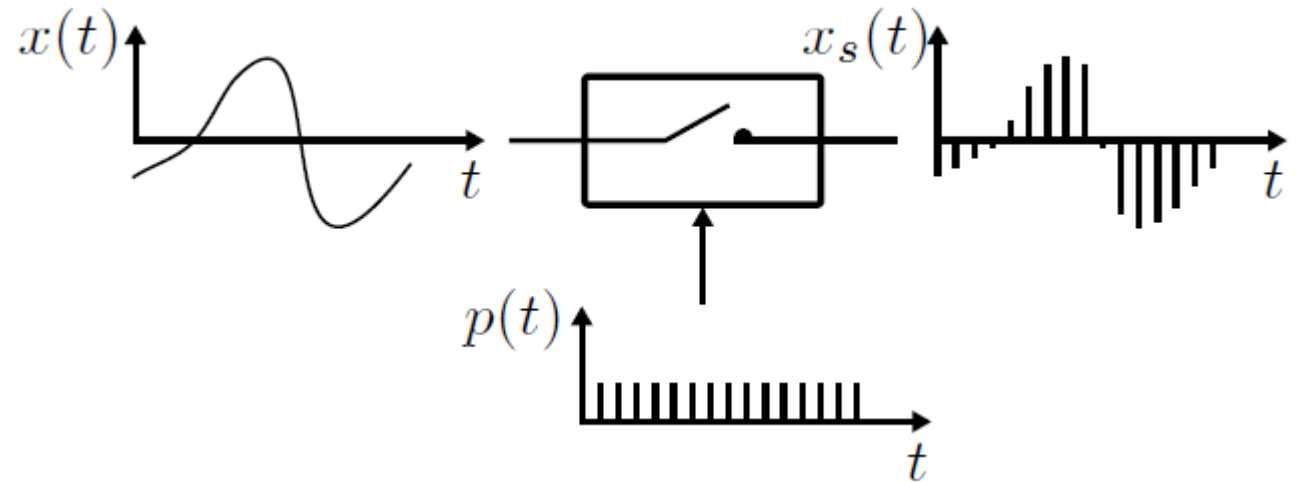
Continuous– discrete (C/D) converter

The input signal to the sampler is the time-continuous signal $x_c(t)$
the sampler's output signal $x_s(t)$ is time-discrete (a periodic pulse train).
The sample switch is controlled by the pulse sampling signal $p(t)$.

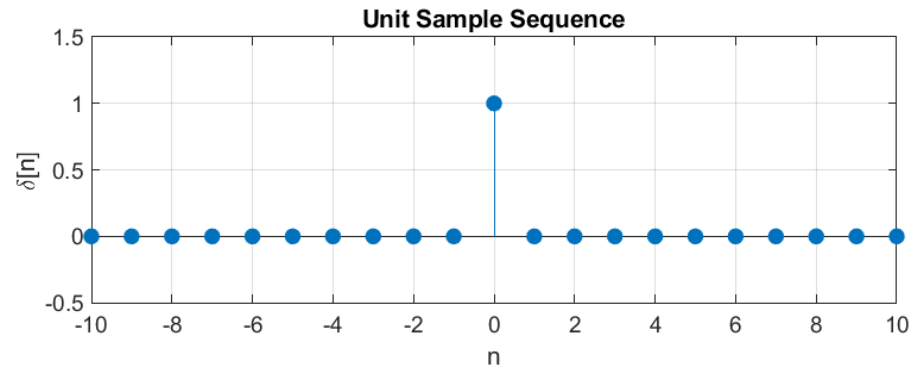
Definition

T : sampling period (interval)

$f_s = \frac{1}{T}$: sampling frequency (Hz)

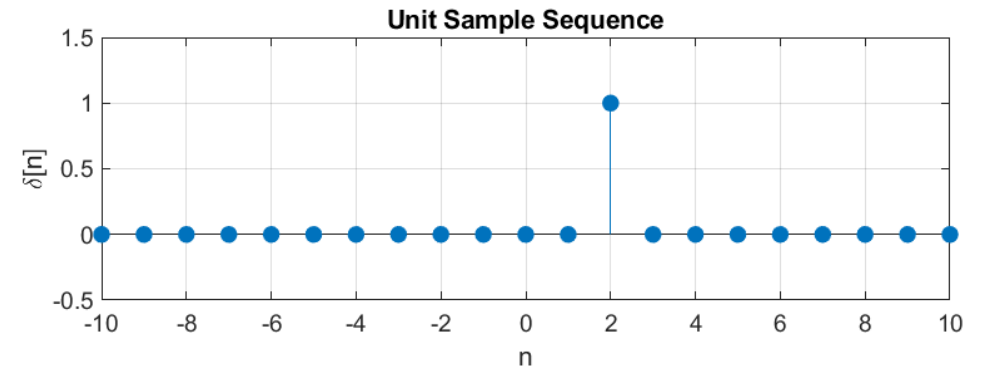


Re-cap



→ Unit sample sequence (impulse)

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases}$$



What is unit sampling at time $t=2$?

Time domain

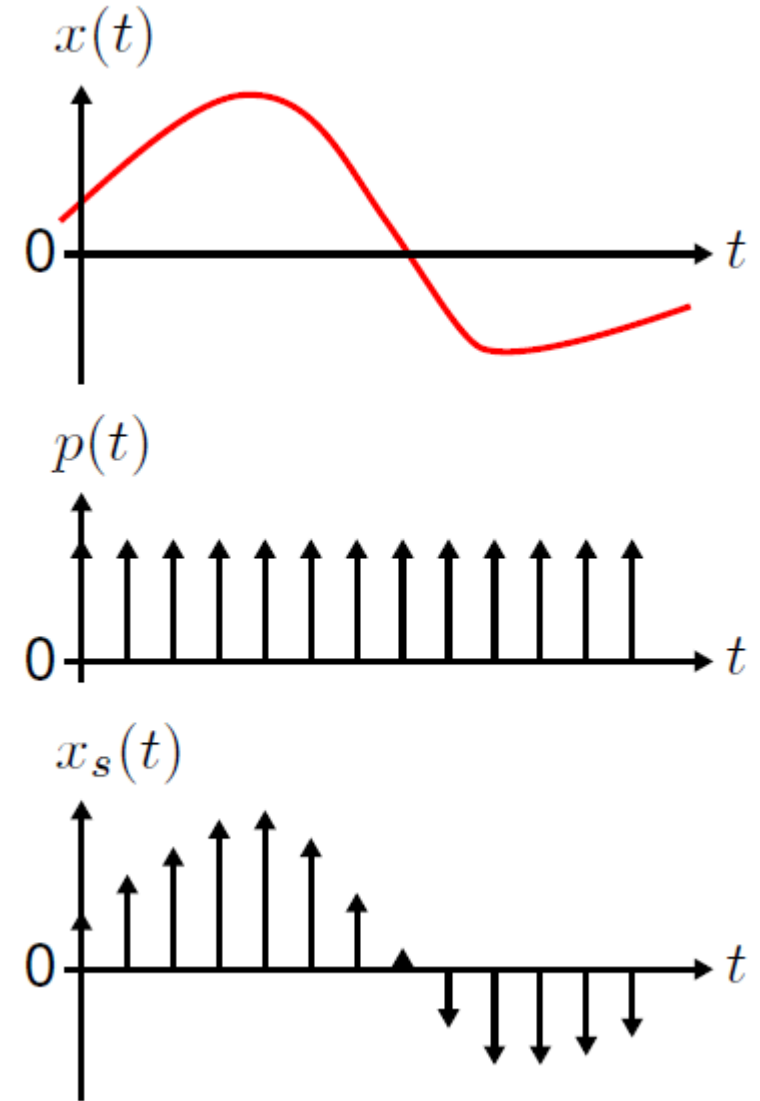
The sampling signal $p(t)$ can be written as
$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

The sampled signal can be described as

$$x_s(t) = x(t)p(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

It can be rewritten as

$$x_s(n) = \sum_{n=-\infty}^{\infty} x(nT) \delta(t - nT)$$



Frequency domain

Applying the above on the pulse sampling signal $p(t)$, we can have

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

Fourier Transform

$$P(f) = \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \delta(t - nT) e^{-j2\pi ft} dt = \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(t - nT) e^{-j2\pi ft} dt = \sum_{n=-\infty}^{\infty} e^{-j2\pi fnT}$$

Sifting property: $\int_{-\infty}^{\infty} f(x) \delta(x - a) dx = f(a)$
--

$$P(f) = \sum_{n=-\infty}^{\infty} e^{-j2\pi fnT} = \frac{1}{T} \sum_{m=-\infty}^{\infty} \delta(f - mf_s)$$

$$P(f) = \sum_{n=-\infty}^{\infty} e^{-j2\pi f n T} = \frac{1}{T} \sum_{m=-\infty}^{\infty} \delta(f - m f_s)$$

The sampled signal $x_s(t)$ can be written as

$$x_s(t) = x(t)p(t)$$

Through Fourier transform

$$X_s(f) = X(f) * P(f) = X(f) * \left[\frac{1}{T} \sum_{m=-\infty}^{\infty} \delta(f - m f_s) \right] = \frac{1}{T} \sum_{m=-\infty}^{\infty} X(f - m f_s)$$

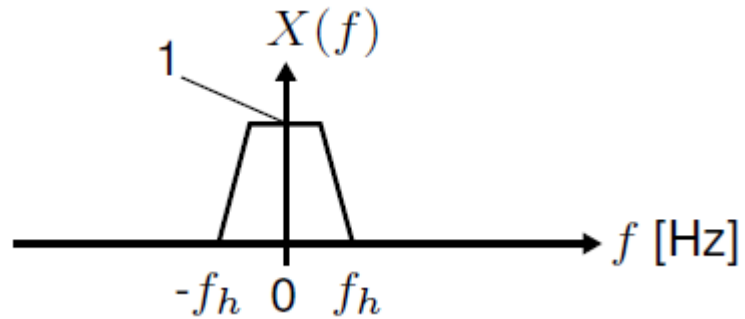
What does this equation mean??

Continuous signal after pulse sampling:

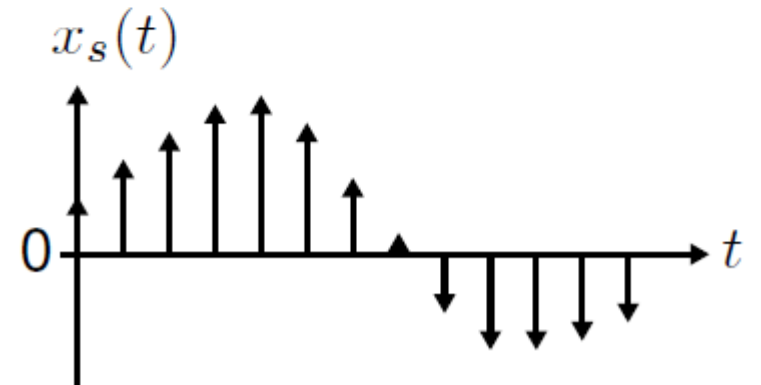
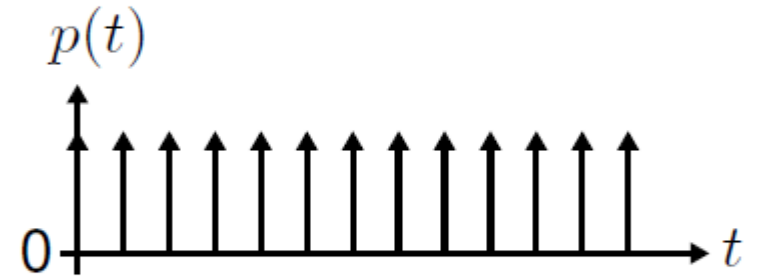
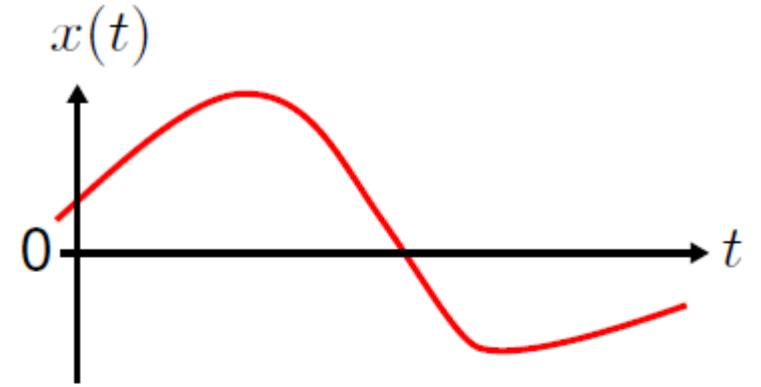
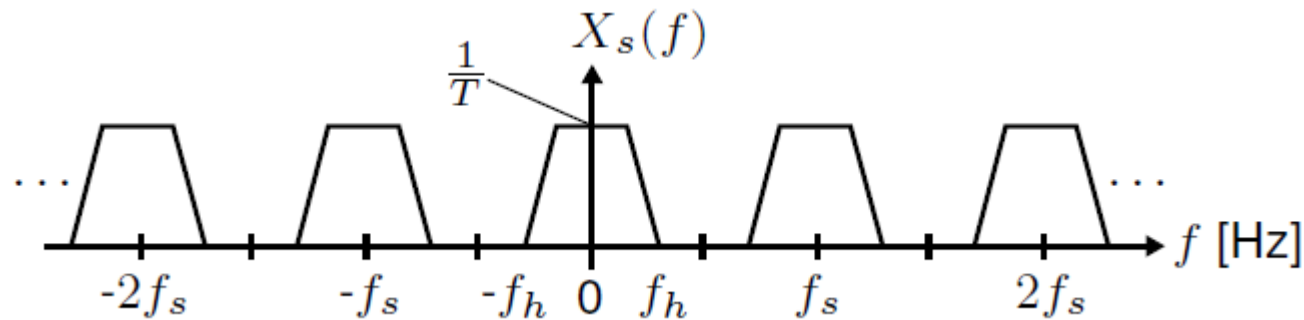
- The original spectrum $X(f)$ is repeated an infinite number of times with distance f_s
- The amplitude is scaled by a factor $1/T$.

Graphical explanation

Example spectrum of continuous signal $X(f)$



After impulse sampling, the amplitude spectrum becomes $X_s(f)$



Relationship btw time domain and freq domain

Discrete signal in time domain $\rightarrow$ periodic frequency spectra

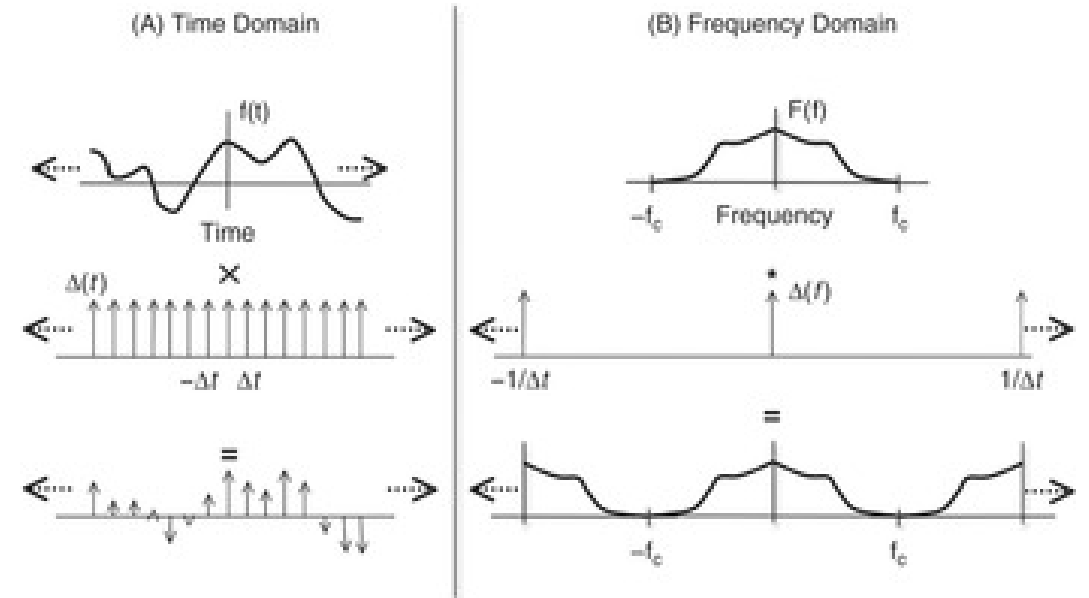
Periodic signal in time domain $\rightarrow$ discrete frequency spectra

A signal that is time-limited $\rightarrow$ it must have infinite spectral content.

A signal that is band-limited $\rightarrow$ it must extend infinitely in time.

Multiply in frequency domain $\rightarrow$ Convolution in time domain

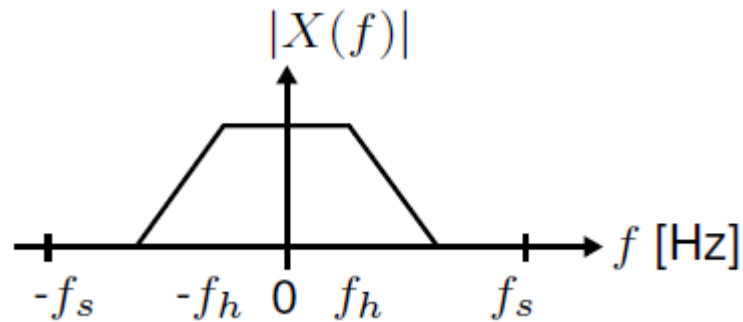
Convolution in frequency domain $\rightarrow$ Multiply in time domain



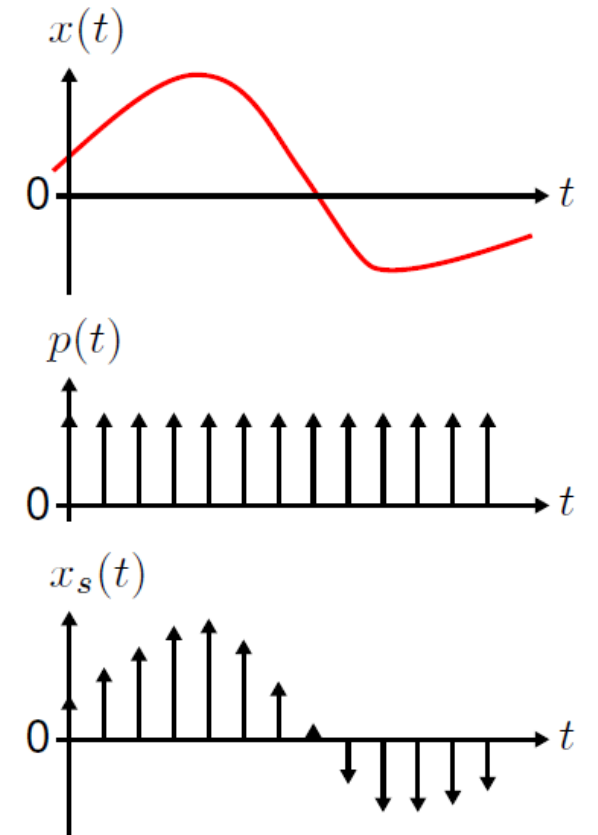
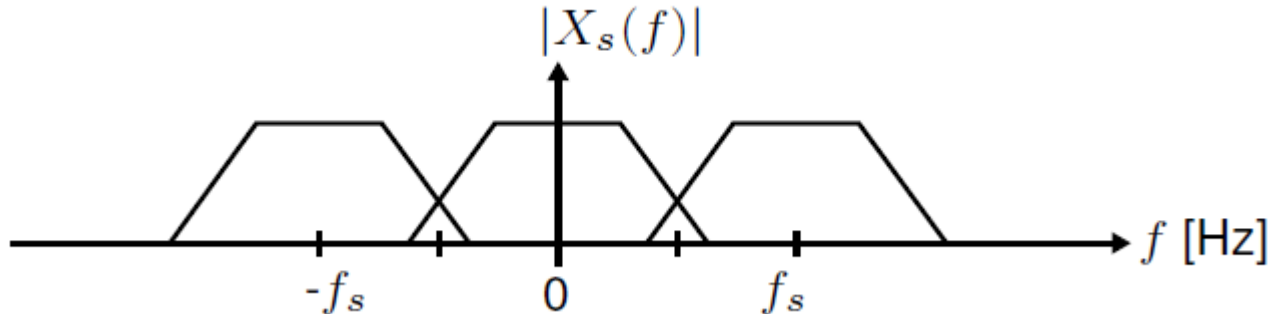
Band limited

How should the sampling frequency f_s be chosen so that the original spectrum can be recovered from the sampled signal?

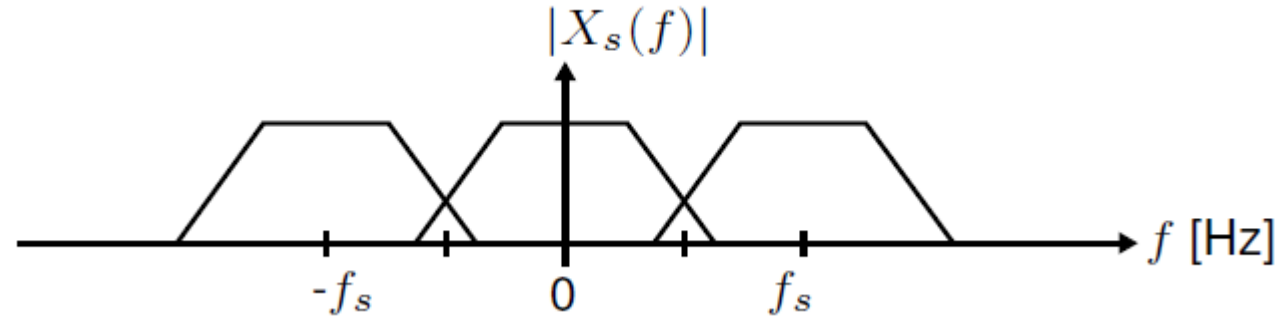
Let's consider the follow example continuous signal spectrum



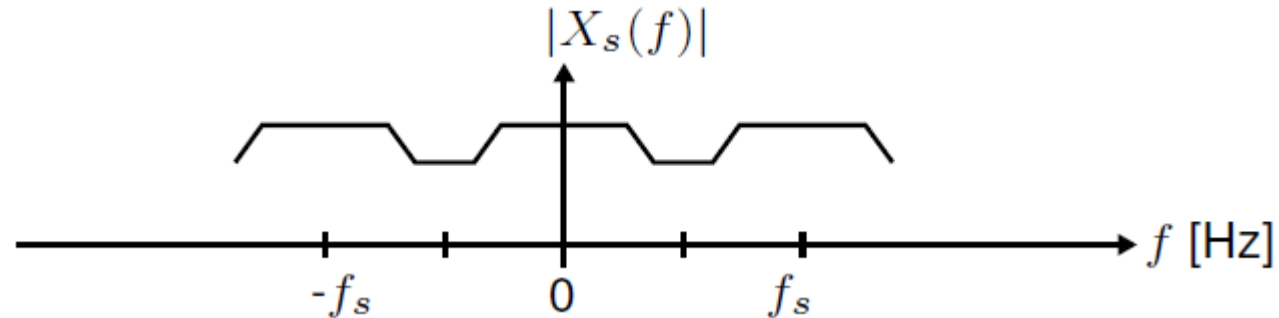
If the sample frequency is chosen too low, the repeated spectra overlap, and thus **aliasing** occurs.



Aliasing



The resulting amplitude spectrum for $x_s(t)$ becomes as below.



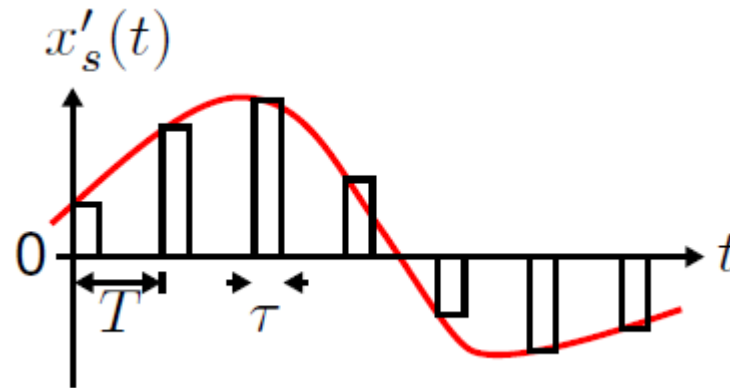
This is called the **aliasing noise**.

Nyquist-Shannon Theorem:

A time-continuous signal $x(t)$ can only be correctly reconstructed from $x_s(t)$ if the sample frequency is at least **twice** the highest frequency in the spectrum for $x(t)$.

Pulse sampling

Impulse sampling cannot be realized in practice, as the pulse width will be greater than zero. In practice, we should consider a pulse sampled signal as shown below.

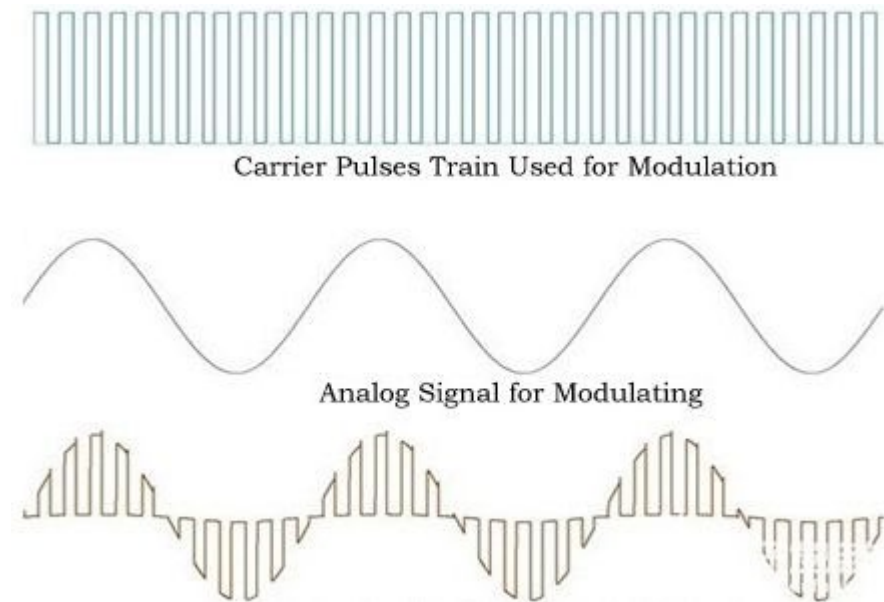


We use T to denote sampling interval [s], and τ to denote the pulse width [s].

The duty factor of the pulse sampled signal $d = \frac{\tau}{T}$

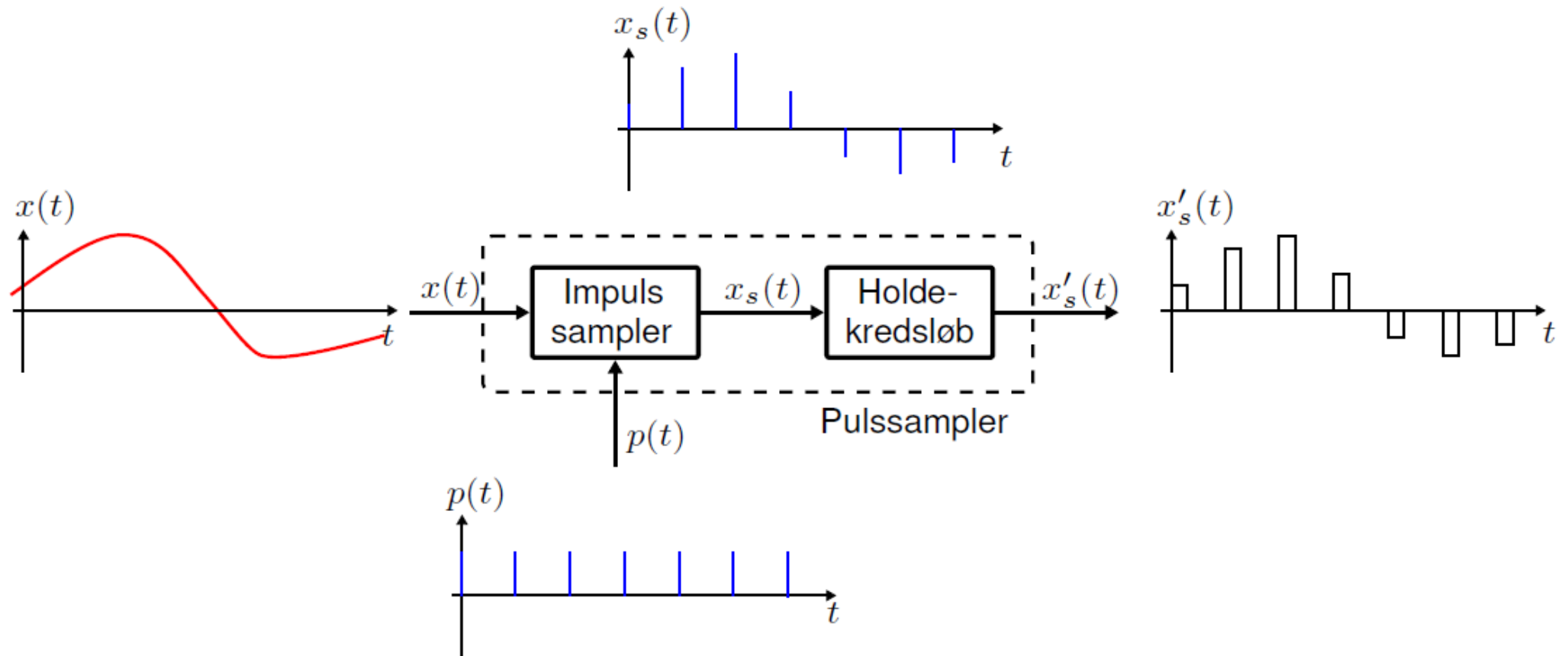
Different types of pulse modulators

- **Pulse Amplitude Modulation**
- Pulse Width Modulation
- Pulse Position Modulation
- Pulse Density Modulation
- etc



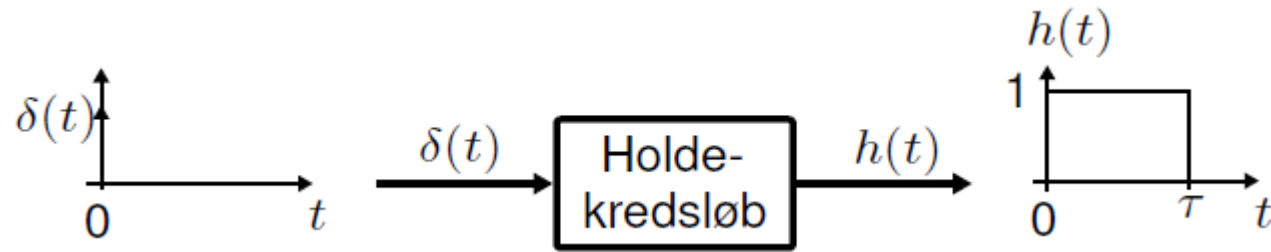
Pulse modulation

A pulse sampler can be modeled using a sample-and-hold circuit



Holding circuit

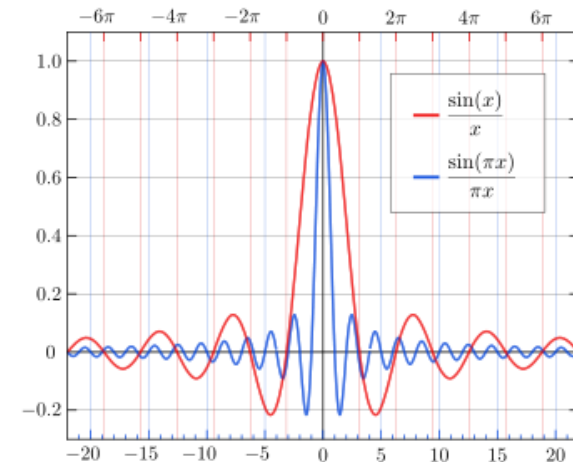
The operation of a pulse sampler with pulse width τ is shown as follows.



It is known that the spectrum function for a **rectangular pulse** is

$$F(\omega) = \mathcal{F}(f(t)) = \tau \operatorname{sinc}\left(\omega \frac{\tau}{2}\right)$$

$$\operatorname{sinc}(x) = \frac{\sin x}{x}$$



Pulse Modulation

$$x_s'(t) = x(t)p(t)f(t)$$

Frequency spectrum of the pulse sampled signal is

$$X_s'(f) = \frac{1}{T} \sum_{m=-\infty}^{\infty} \boxed{X(f)} \boxed{F(f)} \boxed{P(f)} e^{-j\pi d \frac{f}{f_s}}$$

Note: In the original image, the terms $X(f)$, $F(f)$, and $P(f)$ are enclosed in boxes. The term $F(f)$ is red, and $P(f)$ is blue. The term τ is also red.

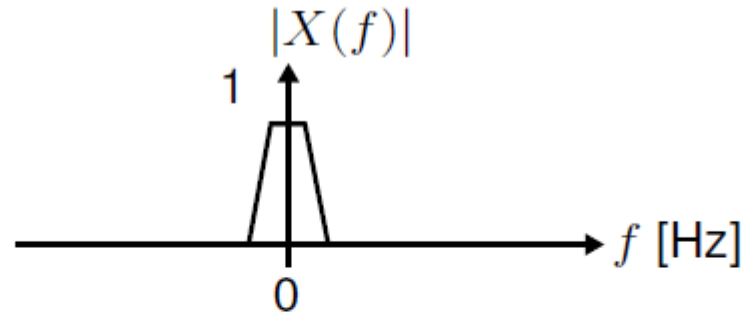
Amplitude spectrum thus becomes

$$|X_s'(f)| = d \left| \text{sinc} \left(\pi d \frac{f}{f_s} \right) \right| \sum_{m=-\infty}^{\infty} X(f - mf_s)$$

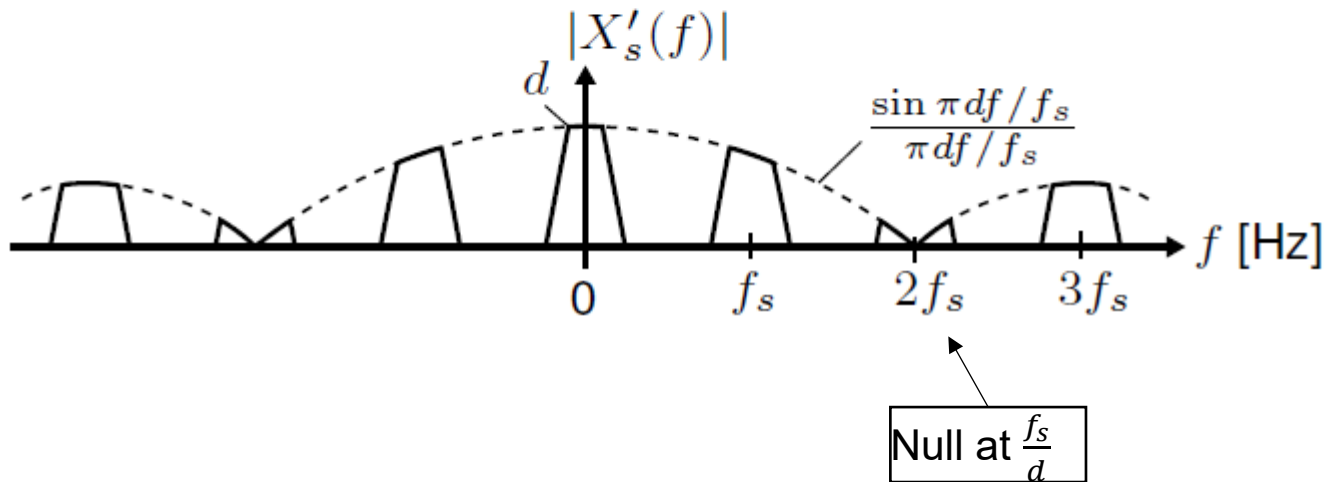
where duty factor of the pulse sampled signal $d = \frac{\tau}{T}$

Pulse-sampling signal with $d = 0.5$

Original
signal



After pulse
sampling



Example

1. Use Matlab to generate a 50Hz + 120Hz sine wave;
2. Impulse sampling at 200Hz, show its amplitude spectrum.
3. Pulse sampling at 200Hz and duty factor 1/2, show its amplitude spectrum.

```
fs = 1000; % "Continuous-time" reference sampling rate (Hz)
T = 1/fs;
t = 0:T:1-T; % 1 second duration
N = length(t);
f = (-N/2:N/2-1)*(fs/N);

% Test Signal (sum of sinusoids)
x = cos(2*pi*50*t) + 0.5*sin(2*pi*120*t); % signal with 50Hz and 120Hz components

% Impulse Sampling
fs_sample = 200; % Sampling frequency (Hz)
Ts = 1/fs_sample;
impulse_train = zeros(size(t));
impulse_train(1:fs/fs_sample:end) = 1; % Dirac comb
x_imp = x .* impulse_train; % Impulse-sampled signal

% Pulse Sampling (with duty factor)
duty = 0.5; % duty cycle (50%)
pulse_train = square(2*pi*fs_sample*t, duty*100);
pulse_train = (pulse_train+1)/2; % Make it 0/1 rectangular pulses
x_pulse = x .* pulse_train; % Pulse-sampled signal

% FFT
X = abs(fftshift(fft(x)))/N;
X_imp = abs(fftshift(fft(x_imp)))/N;
X_pulse = abs(fftshift(fft(x_pulse)))/N;
```

```

% Plot Time Domain
figure;
subplot(3,1,1);
plot(t, x, 'k');
title('Original Signal (Time Domain)');
xlabel('Time (s)'); ylabel('x(t)'); grid on; xlim([0 0.1]);

subplot(3,1,2);
stem(t, x_imp, 'b', 'Marker','none');
title('Impulse Sampled Signal');
xlabel('Time (s)'); ylabel('x_{imp}(t)'); grid on; xlim([0 0.1]);

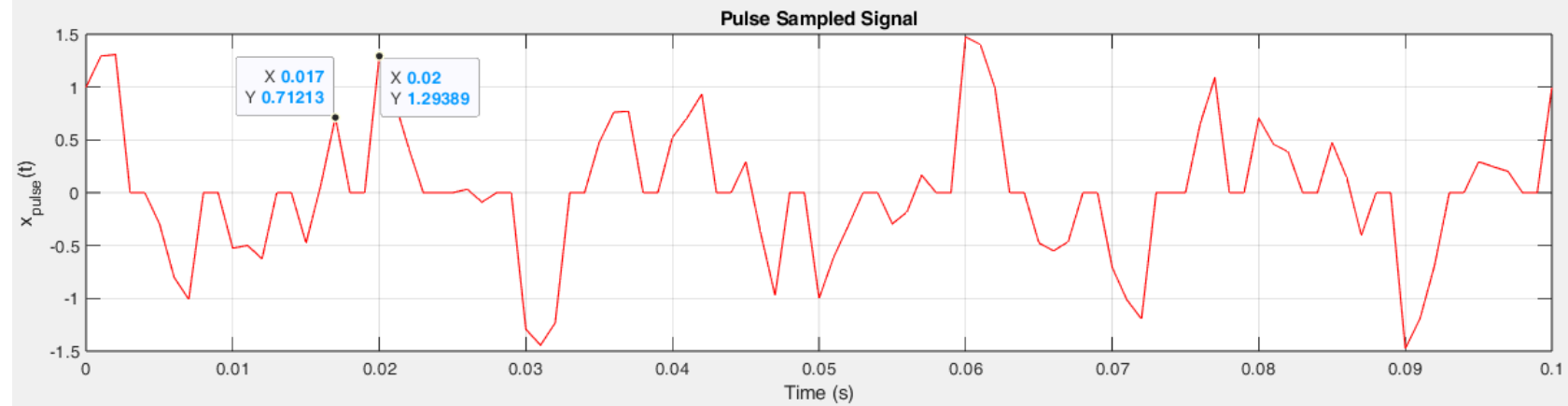
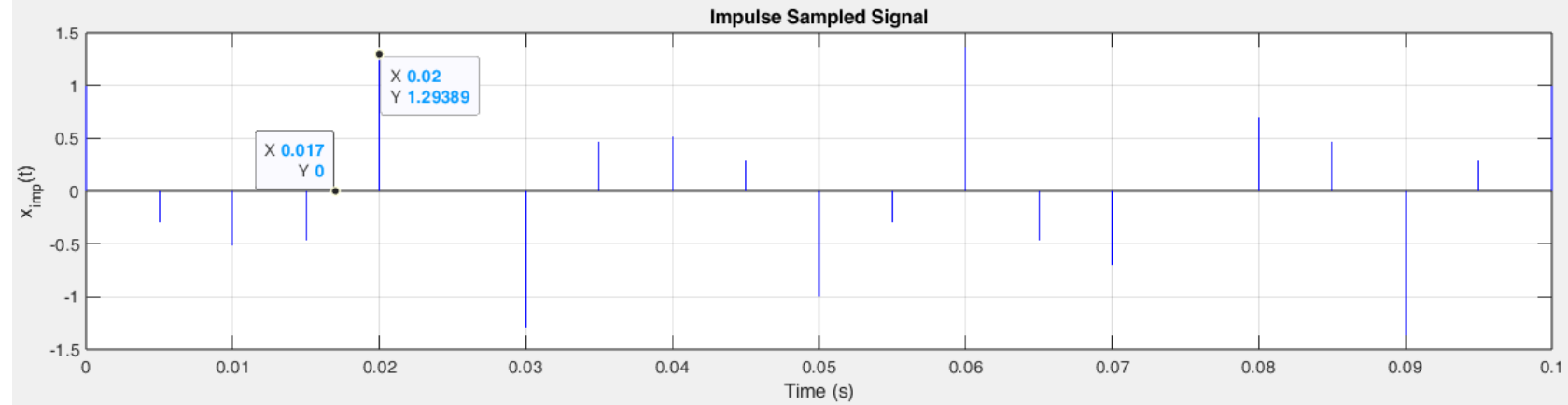
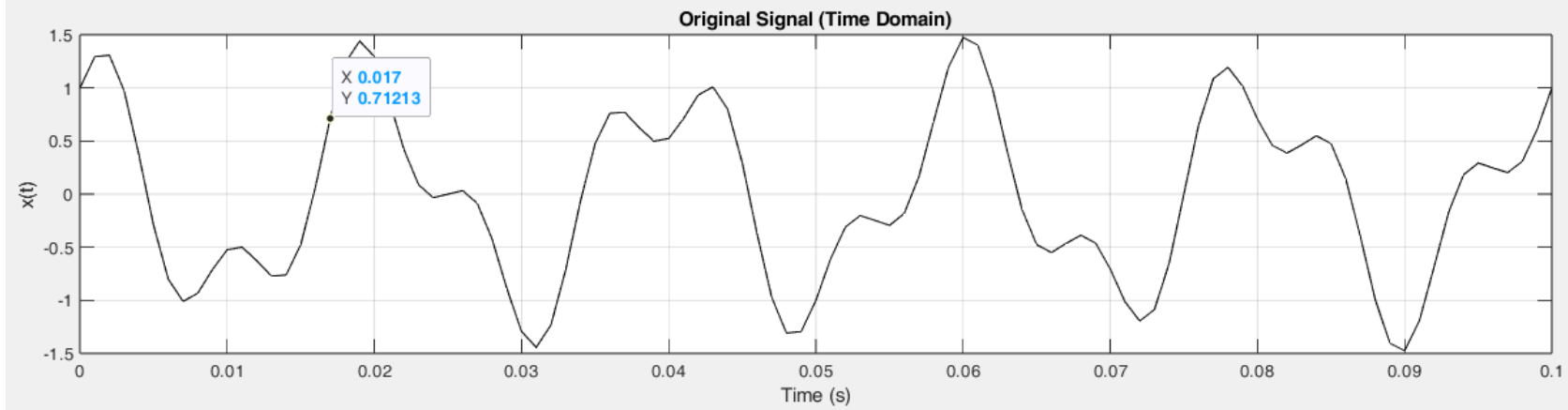
subplot(3,1,3);
plot(t, x_pulse, 'r');
title('Pulse Sampled Signal');
xlabel('Time (s)'); ylabel('x_{pulse}(t)'); grid on; xlim([0 0.1]);

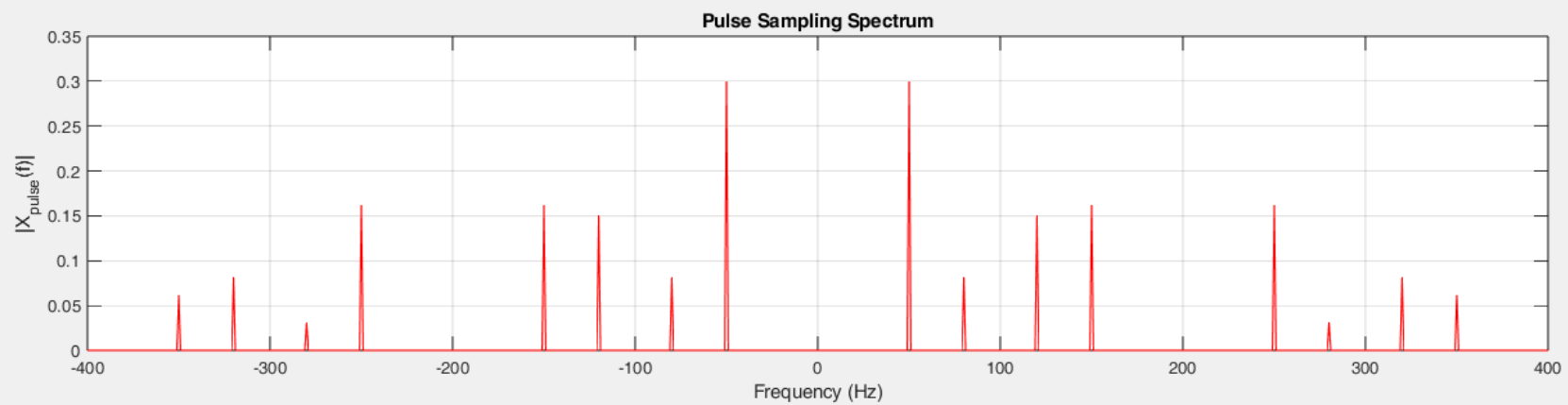
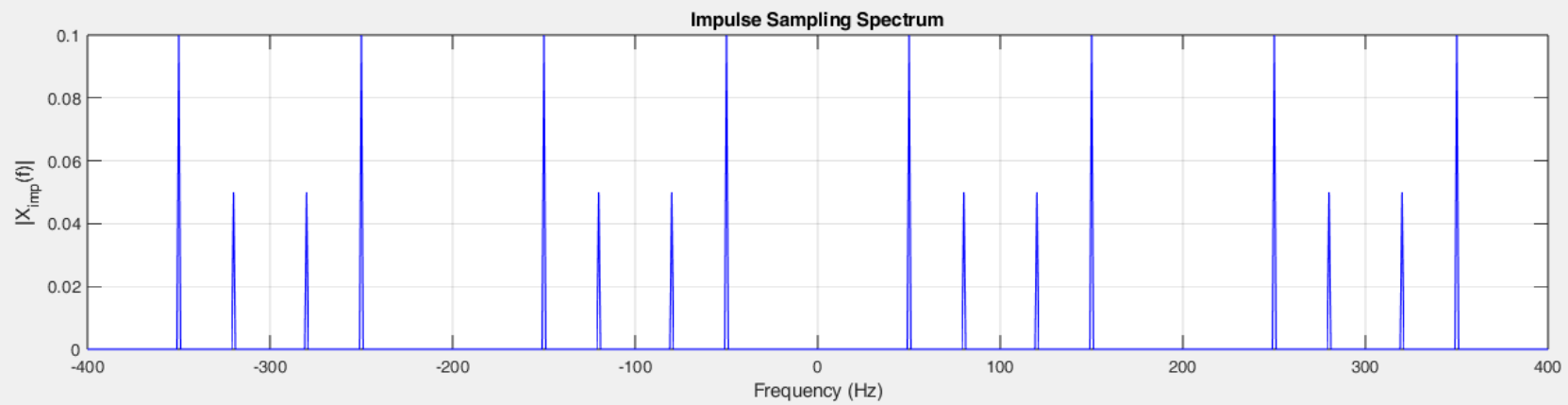
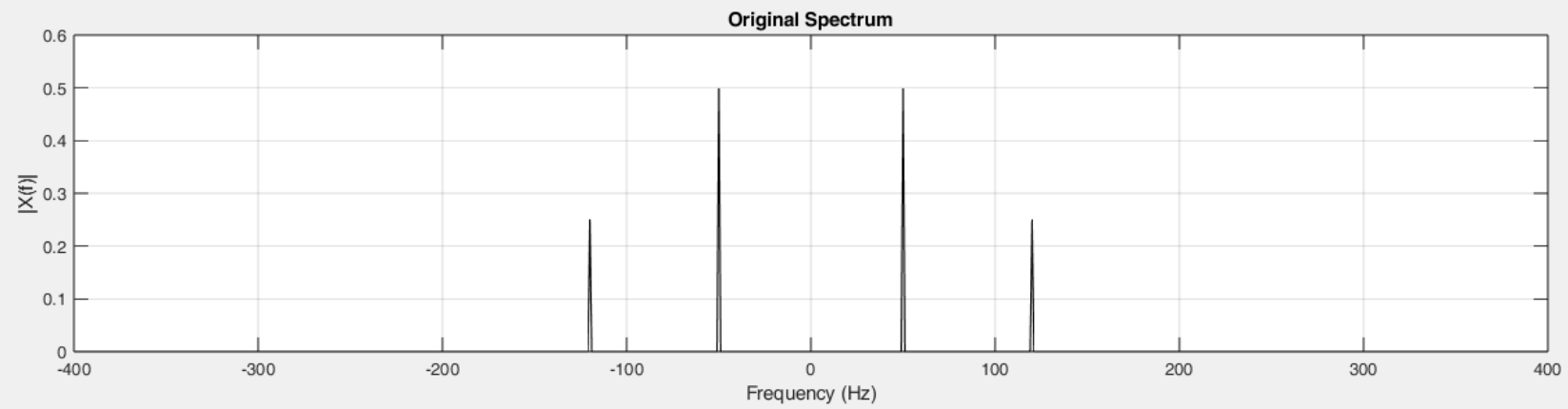
% Plot Frequency Spectrum
figure;
subplot(3,1,1);
plot(f, X, 'k');
title('Original Spectrum');
xlabel('Frequency (Hz)'); ylabel('|X(f)|'); grid on; xlim([-400 400]);

subplot(3,1,2);
plot(f, X_imp, 'b');
title('Impulse Sampling Spectrum');
xlabel('Frequency (Hz)'); ylabel('|X_{imp}(f)|'); grid on; xlim([-400 400]);

subplot(3,1,3);
plot(f, X_pulse, 'r');
title('Pulse Sampling Spectrum');
xlabel('Frequency (Hz)'); ylabel('|X_{pulse}(f)|'); grid on; xlim([-400 400]);

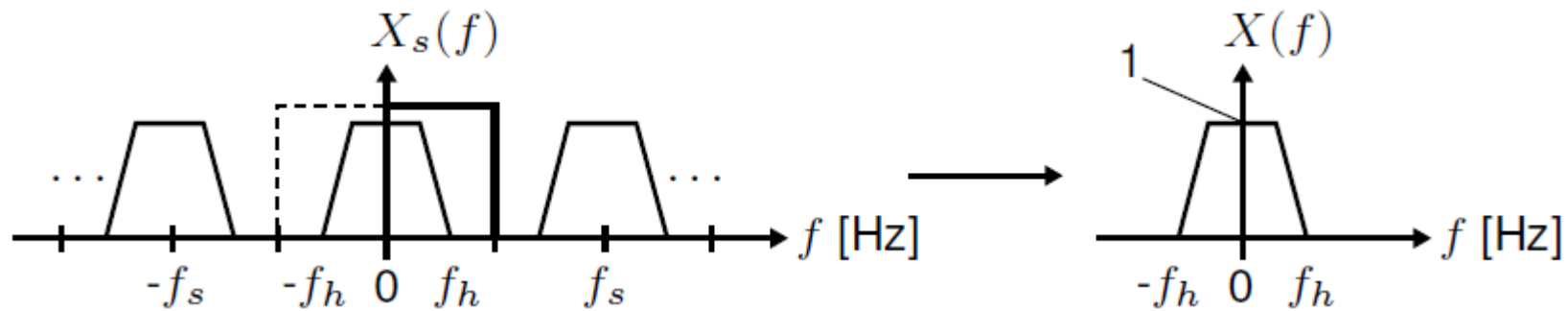
```





Reconstruction

The amplitude spectrum $X(f)$ can be recreated from $X_s(f)$ by **low-pass filtering**.



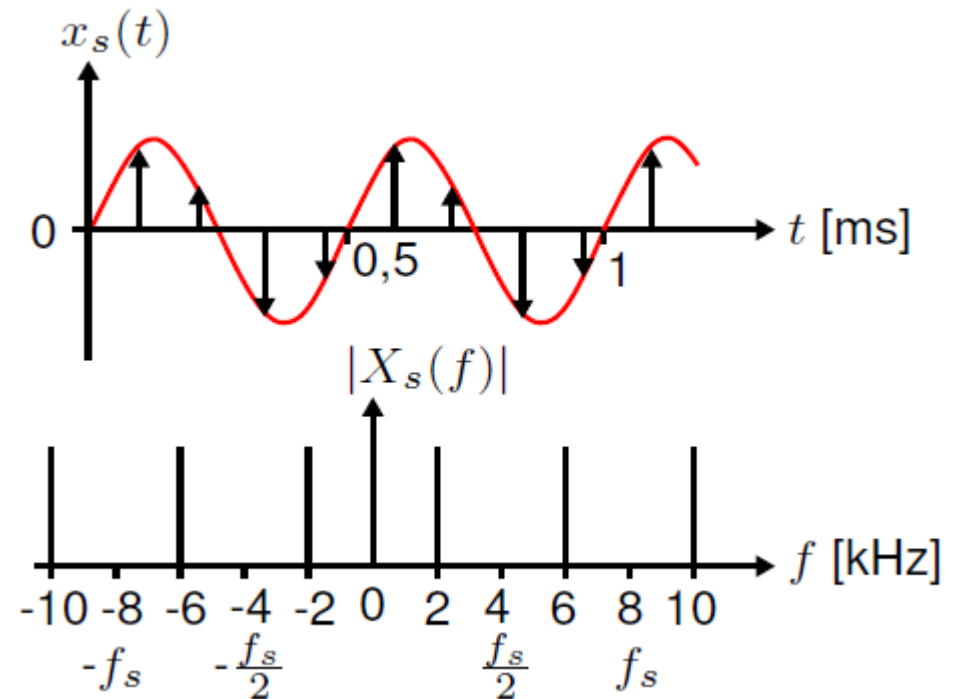
Example

Let's take a look the following signal

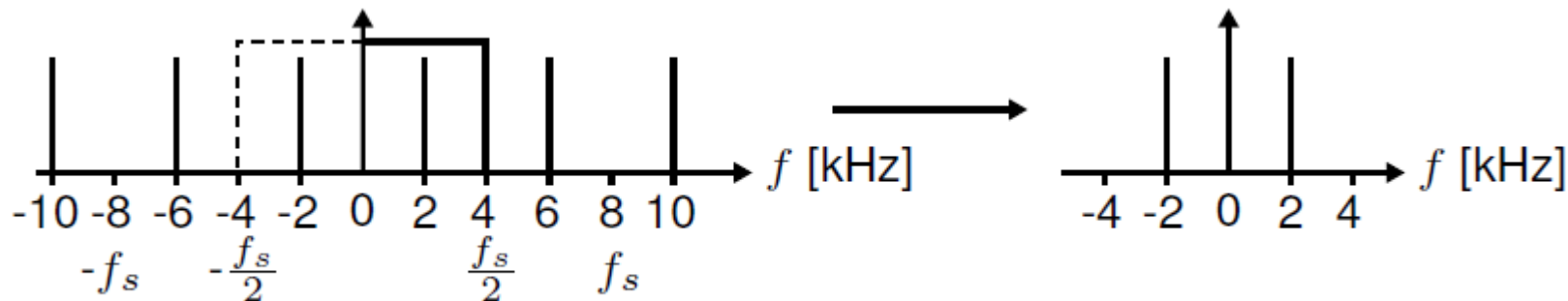
$$x(t) = \sin(2\pi ft)$$

where the signal frequency $f = 2$ kHz

We use a sampling frequency $f_s = 8$ kHz



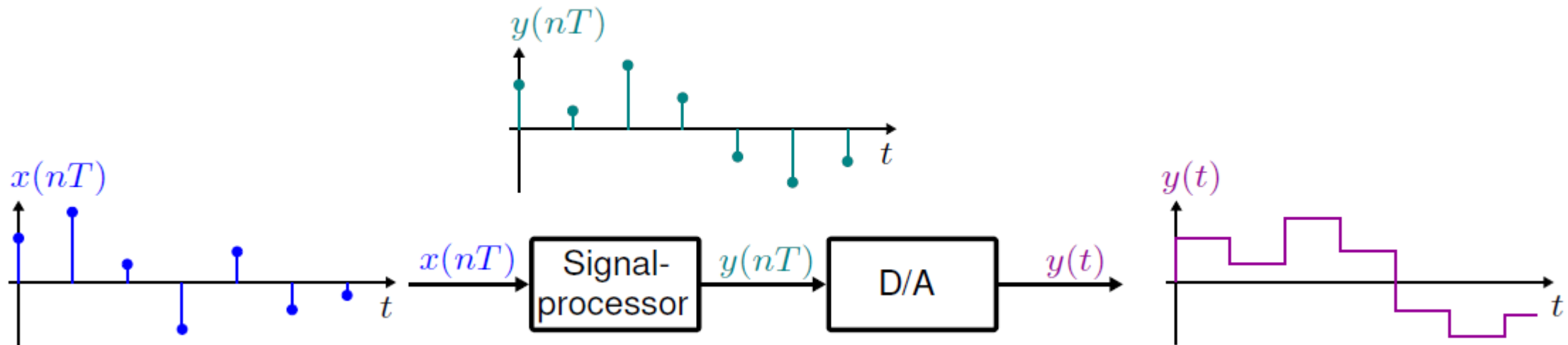
And it can be reconstructed as follows



Reconstruction

Reconstruction means that all the values between 0 and T are interpolated.

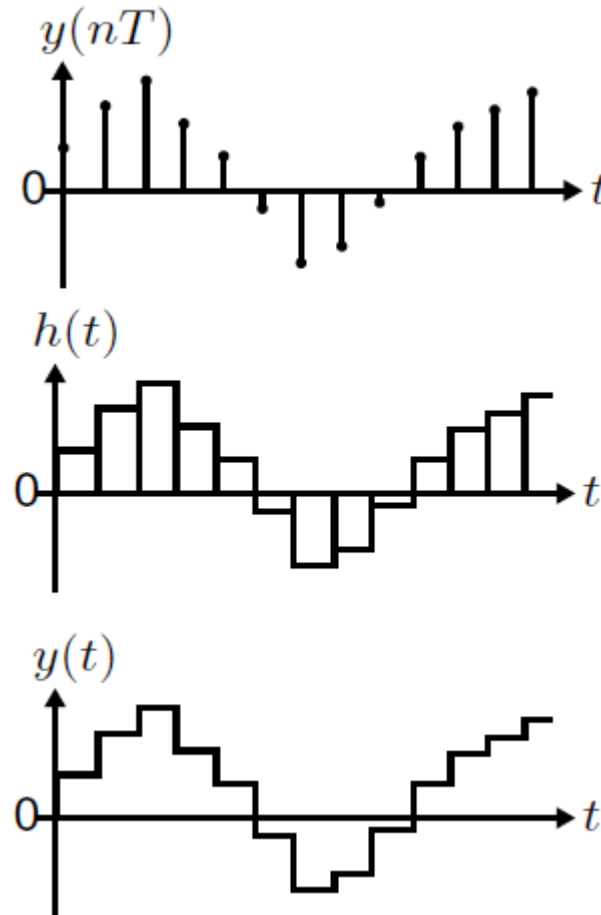
A digital-analog converter (D/A converter) reconstructs the analog signal from the digital sequence, and is therefore called a reconstruction circuit.



Since the output signal of the D/A converter is piecewise **constant** with an amplitude given by the amplitude of the digital sequence, this is a **zero-order hold (ZOH)** circuit.

Zero-order hold circuit

When a digital sequence is D/A converted using zero-order hold circuit acquisition, we get the following

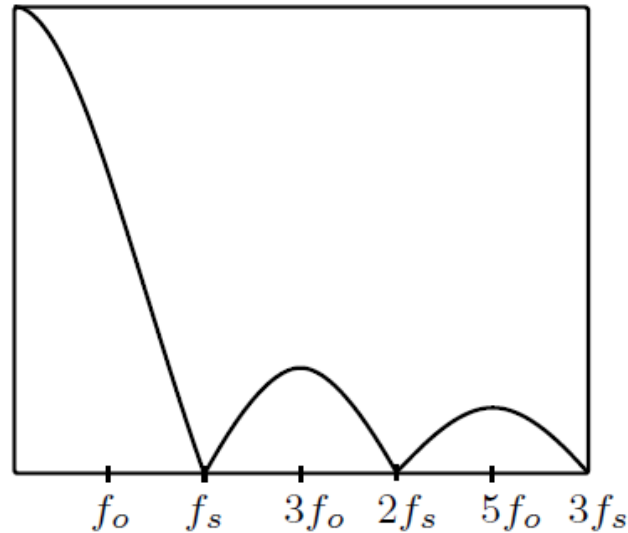


Note that the impulse response of the holding circuit is like that of a pulse sampler, where $\tau = T$ (duty factor of 1).

Amplitude characteristic for zero-order hold circuit

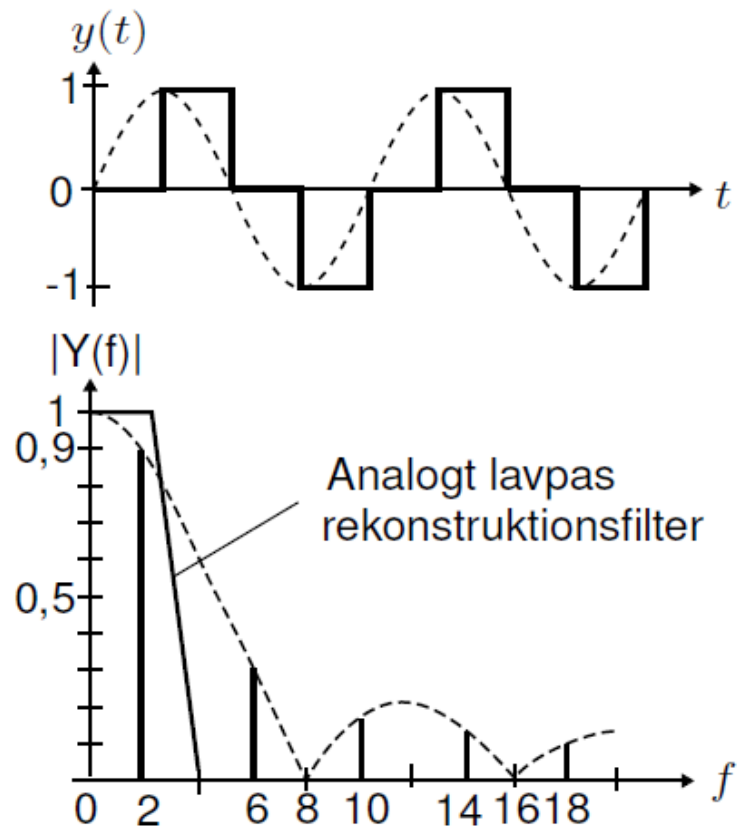
The frequency response of a zero-order holding circuit with input signal $x(t)$ and output signal $y(t)$ is

$$|Y(f)| = \left| \frac{\sin\left(\frac{\pi f}{f_s}\right)}{\frac{\pi f}{f_s}} \right| \sum_{m=-\infty}^{\infty} X(f - mf_s)$$



Example

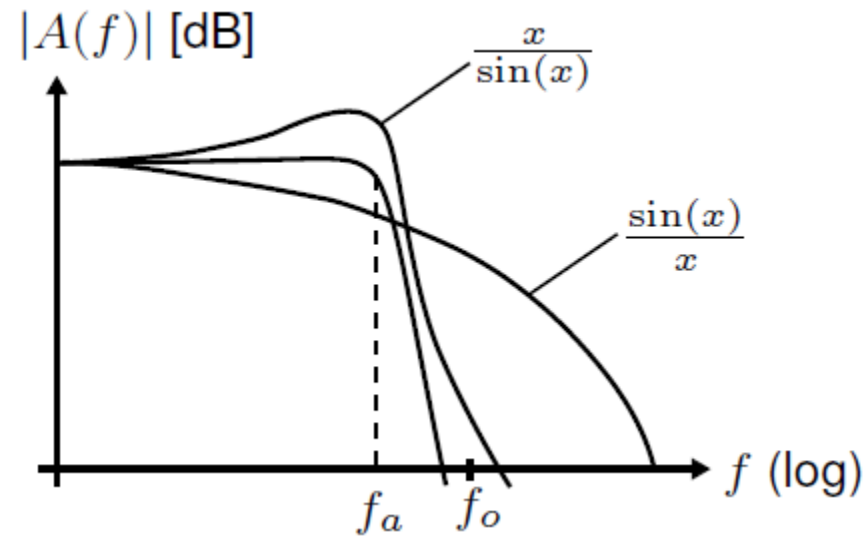
The following shows the reconstruction of a 2 kHz sine wave sampled at 8 kHz.



A low-pass filter is often inserted after the D/A converter to attenuate the high-frequency components at $f > f_o$.

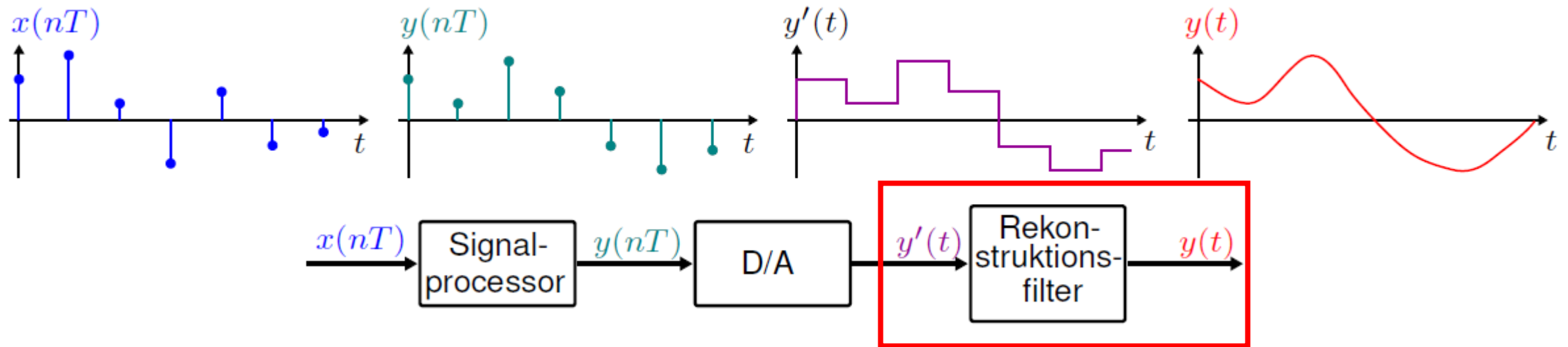
Correction of amplitude response

Since the $\text{sinc}()$ function is multiplied on the amplitude spectrum of the signal, a reconstruction filter with the reciprocal amplitude characteristic $\frac{x}{\sin(x)}$ can be added.



Reconstruction filter

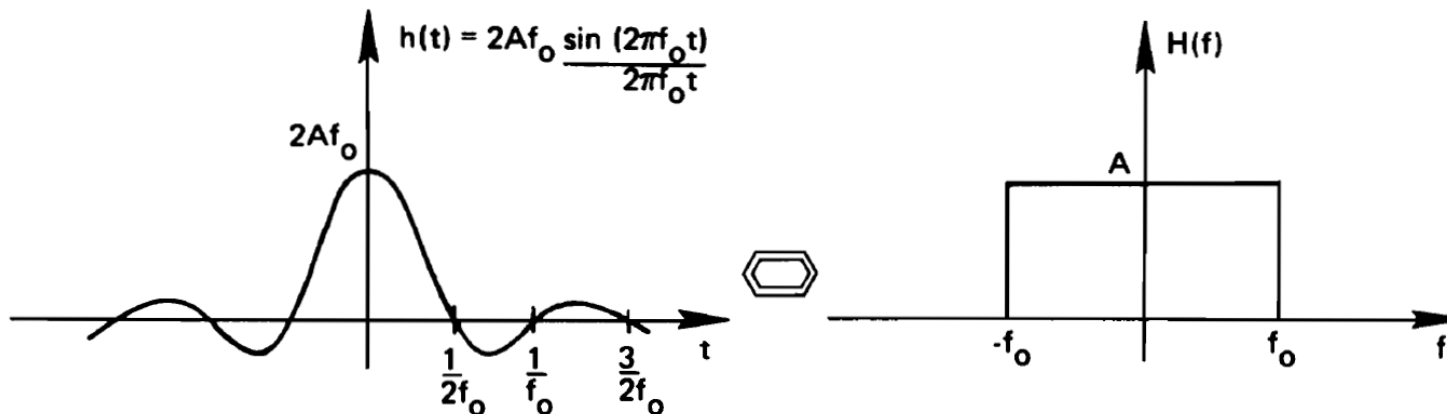
Reconstruction filter: to remove frequency components with frequency above f_o , a low-pass filter is used.



Reconstruction in the time domain

$$\begin{aligned}x_r(t) &= h_r(t) * x_s(t) = h_r(t) * \sum_n x(n) \delta_c(t - nT) = \sum_n x(n) h_r(t - nT) \\&= \dots + x(-1)h_r(t + T) + x(0)h_r(t) + x(1)h_r(t - T) + \dots\end{aligned}$$

where $h_r(t)$ is called the reconstruction filter



Multiply in frequency domain
Means
Convolution in time domain

Example - reconstruction

- Generate an original signal 1Hz sin wave of 4s.
- Do an impulse sample of 28 samples
- Present the zero-order hold result
- Reconstruct analog signal from the discrete time domain samples

```
clc; clear; close all;

t = 0:0.001:4-0.001;
x_org = sin(2*pi*t);

t_smp = linspace(0, 4, 29); % 28+1
f_s = 28/4; % sampling freq
x_n = sin(2*pi*t_smp);
figure();
plot(t,x_org)
hold on
stem(t_smp, x_n)

% Zero-order hold (ZOH)
t_hold = linspace(0, 4, 1001); % Continuous time vector for holding
y_zoh = interp1(t_smp, x_n, t_hold, 'previous'); % ZOH interpolation

% Sinc Interpolation for Reconstruction
T = 1/f_s;
y_rec = zeros(size(t_hold)); % Initialize reconstructed signal
for k = 1:length(t_smp)-1
    y_rec = y_rec + x_n(k) * sinc((t_hold - t_smp(k)) / T);
end

figure()
subplot(2,1,1)
plot(t_hold, y_zoh)

subplot(2,1,2)
plot(t_hold, y_rec)
```

2.3 ANALOG-TO-DIGITAL CONVERSION, DIGITAL-TO-ANALOG CONVERSION, AND QUANTIZATION

Reference book
(Page 35-47)

11.1 LINEAR MIDTREAD QUANTIZATION

Reference book
(Page 497-500)

Signal noise ratio (SNR)

The SNR is defined as the ratio of signal power to noise power

$$SNR = \frac{E(x^2)}{E(e_q^2)}$$

Where x is the signal, and e_q is the noise.

In practice, the SNR can be calculated as

$$SNR = \frac{\sum_{n=0}^{N-1} x^2(n)}{\sum_{n=0}^{N-1} e_q^2(n)}$$

SNR can also be represented in dB

$$SNR_{dB} = 10 \log_{10}(SNR) \text{ dB}$$

Example: SNR

Given expected signal x as [2, -2, 2, -2], while the actually measured signal y as [2.5, -2.5, 2.5, -2.5]

So, noise e_q is [0.5, -0.5, 0.5, -0.5]

The signal power

$$P_x = \frac{1}{N} \sum_{n=0}^{N-1} x^2(n) = \frac{2^2 + (-2)^2 + 2^2 + (-2)^2}{4} = 4$$

The noise power

$$P_n = \frac{1}{N} \sum_{n=0}^{N-1} e_q^2(n) = \frac{0.5^2 + (-0.5)^2 + 0.5^2 + (-0.5)^2}{4} = 0.25$$

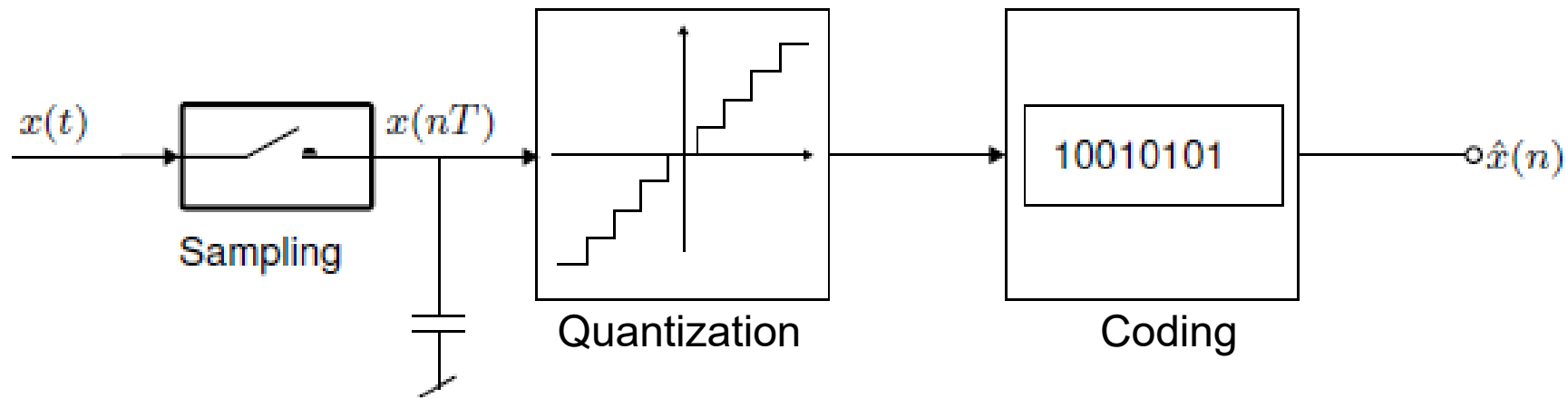
The SNR is

$$SNR = \frac{P_x}{P_n} = \frac{4}{0.25} = 16$$

It can also be written

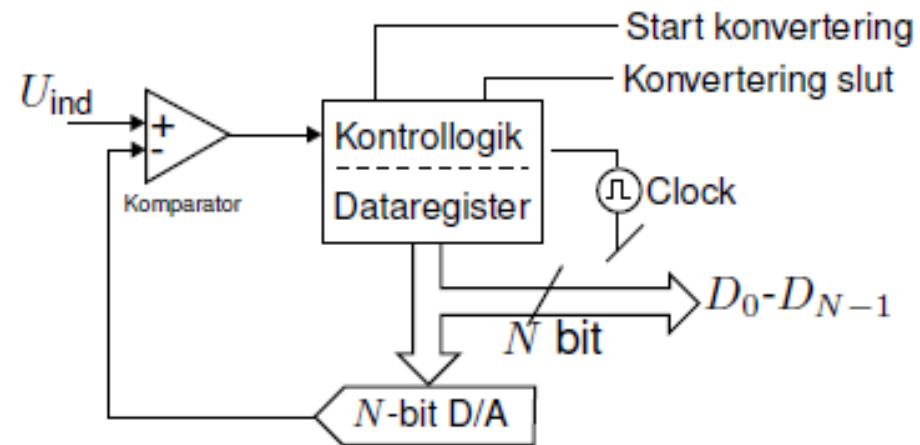
Quantization

In analog-digital conversion, the input signal $x(t)$ is transformed into a sequence with finite resolution as shown below.



Analog-Digital converter principle

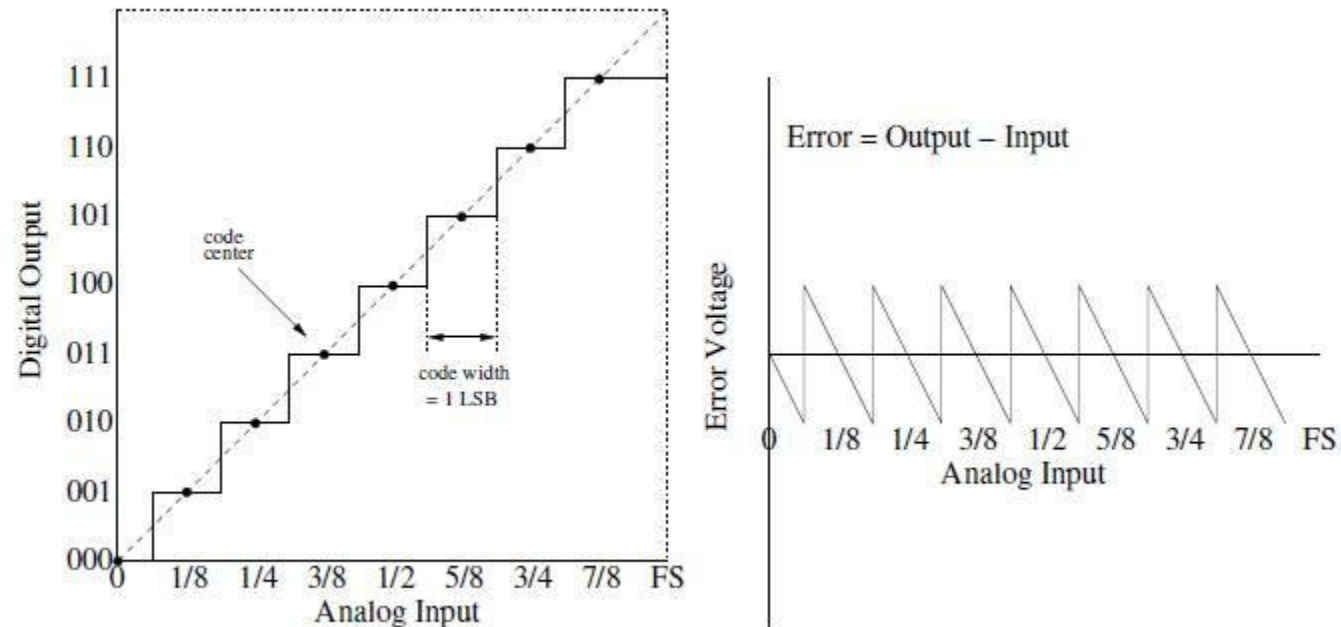
A Successive Approximation A/D converter is built according to the principle shown below, where the output of the A/D converter is compared with the input sequence through a comparator.



If the conversion is to take place faster (in the MHz range), then a Flash A/D converter is used.

Quantization

During A/D conversion, the input signal is quantized so that it has a word length of N bits, i.e. the representation has 2^N different quantization levels.



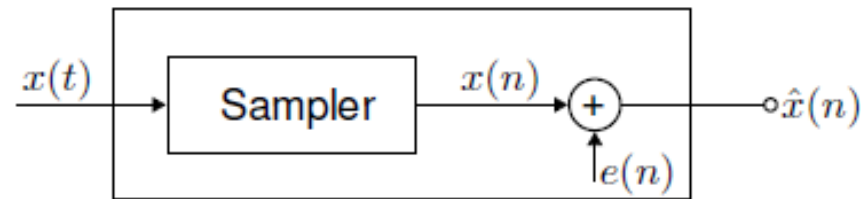
Quantization error

The quantization error means that the value of the input sequence $x(n)$ and the quantized input sequence $\hat{x}(n)$ are different

$$\hat{x}(n) = x(n) + e(n)$$

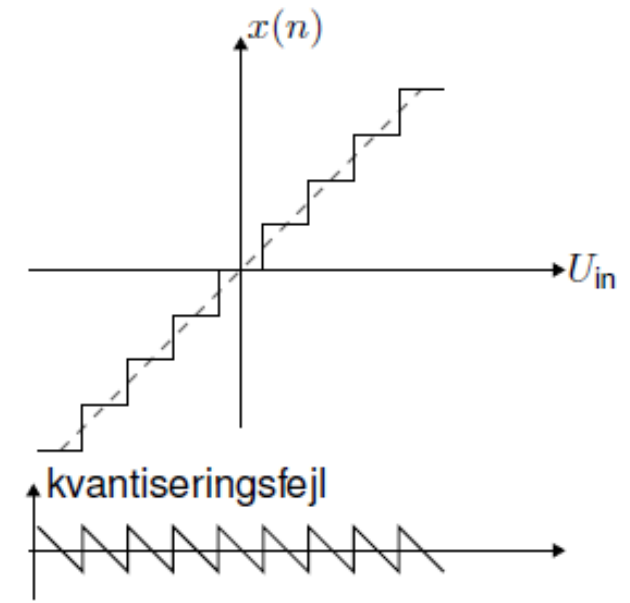
where $e(n)$ is the quantization error.

The quantization error can be thought of as a noise sequence as shown below.



The magnitude of the quantization error e_q is bounded as

$$-\frac{\Delta V}{2} \leq e_q \leq \frac{\Delta V}{2}$$



Quantization error

The signal varies from $-V/2$ to $+V/2$, and has 2^N quantization levels.

$$V = 2^N \Delta V$$

The quantization error is modeled as a uniform distribution in $-\frac{\Delta V}{2} \leq e_q \leq \frac{\Delta V}{2}$. According to the theory of probability and random variables, the quantization noise is calculated as

$$E(e_q^2) = \frac{\Delta V^2}{12}$$

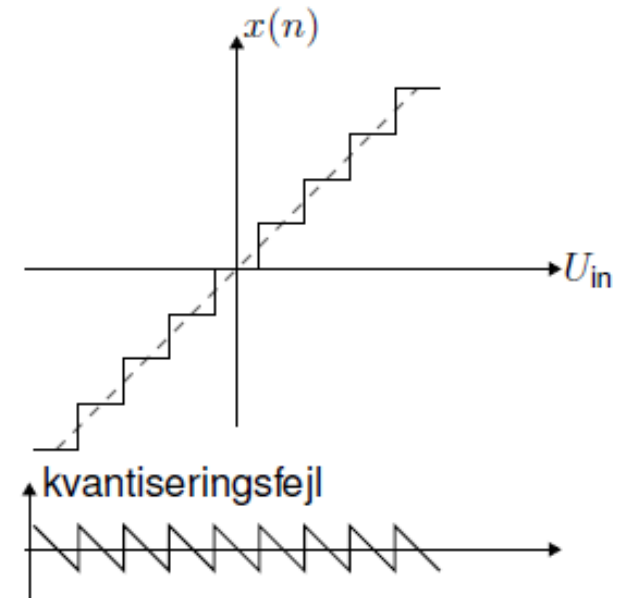
For linear signal input, Signal-to-Quantization Noise Ratio (SQNR) can be calculated as

$$SQNR = \frac{E(x^2)}{E(e_q^2)} = 2^{2N}$$

$$SQNR_{dB} = 10 \log_{10}(2^{2N}) \approx 6.02 N \text{ [dB]}$$

For full-scale sinusoidal signal input, SQRT can be calculated as

$$SQNR_{dB} = 1.76 + 6.02 N \text{ [dB]}$$



Exercise

2.5. Given an analog signal

$$x(t) = 5\cos(2\pi \cdot 2,500t) + 2\cos(2\pi \cdot 4,500t), \text{ for } t \geq 0$$

sampled at a rate of 8,000 Hz,

- a. sketch the spectrum of the sampled signal up to 20 kHz;
- b. sketch the recovered analog signal spectrum if an ideal lowpass filter with a cutoff frequency of 4 kHz is used to filter the sampled signal in order to recover the original signal;
- c. determine the frequency/frequencies of aliasing noise.

Exercise

- 2.25. Assuming that a 4-bit ADC channel accepts analog input ranging from 0 to 5 volts, determine the following:
- a. number of quantization levels;
 - b. step size of the quantizer or resolution;
 - c. quantization level when the analog voltage is 3.2 volts;
 - d. binary code produced by the ADC;
 - e. quantization error.

Exercise

Use the MATLAB programs in Section 2.5 to solve the following problems.

2.31. Given a sinusoidal waveform of 100 Hz,

$$x(t) = 4.5\sin(2\pi \times 100t)$$

sample it at 8,000 samples per second and

- a. write a MATLAB program to quantize $x(t)$ using a 6-bit bipolar quantizer to obtain the quantized signal x_q , assuming that the signal range is from -5 to 5 volts;
- b. plot the original signal and quantized signal;
- c. calculate the SNR due to quantization using the MATLAB program.